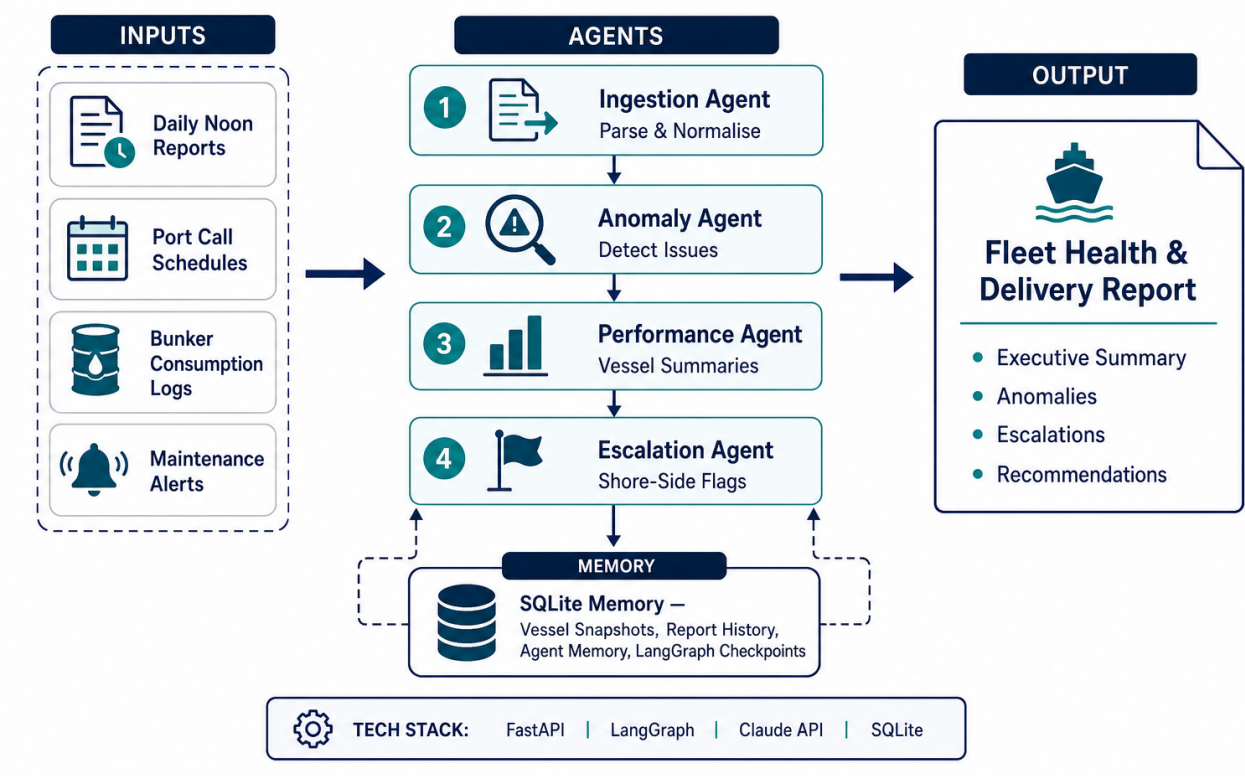


Fleet Health & Delivery Report

Architecture Write-Up - ThinkPalm AgentAI Team Alpha

Fleet Health & Delivery Report – Multi-Agent Pipeline



Problem

Ship management teams receive daily data from noon reports, port schedules, bunker logs, and maintenance alerts. Manual synthesis is slow and error-prone.

Solution

A sequential 4-agent LangGraph pipeline with Claude API tool-calling produces structured Fleet Health & Delivery Reports via FastAPI.

Agents

1. Ingestion - parse and normalise data
2. Anomaly - detect fuel, schedule, maintenance issues
3. Performance - draft vessel summaries for superintendent
4. Escalation - flag critical shore-side defects

Memory

SQLite stores vessel snapshots, report history, agent memory, and LangGraph checkpoints.

Stack

FastAPI, LangGraph, Claude API (Anthropic), SQLite, Pydantic, pytest